ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

ELECTRICAL TECHNOLOGY

THEORY PAPER

MARKS: 100 **TIME: 2 hours**

Topics: occupational health and safety, single-phase AC and RLC circuits, semiconductor devices and power supplies, logic and switching circuits. Suitable for Electronics, Digital and Power Systems learners.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL formulae, substitutions and units in calculations. Round answers to TWO decimal places.
2. A bar over a letter (e.g. $\overline{A}$) means NOT A. A non-programmable calculator may be used.

QUESTION 1: MULTIPLE CHOICE

1.1 In a purely inductive AC circuit, the current …

A leads the voltage by 90°. B lags the voltage by 90°.
C is in phase with the voltage. D is zero.

1.2 At resonance in a series RLC circuit …

A $X_L = X_C$ and the impedance is a minimum. B $X_L = X_C$ and the current is a minimum.
C $X_L = R$. D the power factor is zero.

1.3 The RMS value of a sine wave with a peak value of 325 V is about …

A 230 V. B 325 V.
C 460 V. D 163 V.

1.4 A transistor used as a switch works in the …

A active region only. B cut-off and saturation regions.
C breakdown region. D reverse-bias region only.

1.5 The Boolean expression for a two-input NAND gate is …

A $Y = A \cdot B$ B $Y = A + B$
C $Y = \overline{A \cdot B}$ D $Y = A \oplus B$

(5 × 2)
[10]

QUESTION 2: OCCUPATIONAL HEALTH AND SAFETY

2.1 Distinguish between an *unsafe act* and an *unsafe condition*. Give ONE example of each in an electrical workshop. (4)

2.2 Explain why you must never work on a live circuit, and state TWO steps of the lock-out procedure. (3)

2.3 State THREE effects that an electric shock can have on the human body. (3)

[10]

QUESTION 3: SINGLE-PHASE AC AND RLC CIRCUITS

3.1 An alternating voltage is given by v = 340 sin (314 t) V. Determine: (a) the peak voltage; (b) the frequency; (c) the RMS voltage; (d) the instantaneous voltage when t = 5 ms. (7)

3.2 A series RLC circuit has a resistor of 30 Ω, an inductor of 0,2 H and a capacitor of 100 µF connected to a 220 V, 50 Hz supply. Calculate:

3.2.1 The inductive reactance (3)

3.2.2 The capacitive reactance (3)

3.2.3 The impedance of the circuit (3)

3.2.4 The current drawn from the supply (3)

3.2.5 The phase angle, and state whether the current leads or lags the supply voltage. Give a reason. (3)

3.2.6 The voltage across the inductor is about 320 V, which is more than the supply voltage. Explain how this is possible. (2)

3.3 Calculate the resonant frequency of the circuit in 3.2. (3)

3.4 State what happens to the impedance, the current and the power factor of a series RLC circuit at resonance. (3)

[30]

QUESTION 4: SEMICONDUCTOR DEVICES AND POWER SUPPLIES

4.1 Explain what happens in a silicon PN-junction diode when it is (a) forward biased and (b) reverse biased. (4)

4.2 Describe how a bridge rectifier with four diodes changes AC to DC during the positive and negative half-cycles, and explain why a capacitor is connected across the load. (6)

4.3 A Zener regulator has a 15 V supply, a series resistor of 220 Ω, a 9,1 V Zener diode and a 1 kΩ load resistor. Calculate: (a) the voltage across the series resistor; (b) the current through the series resistor; (c) the load current; (d) the Zener current. (8)

4.4 A transistor has a current gain (β) of 100 and a base current of 40 µA. Calculate the collector current and the emitter current. (4)

4.5 An LED (2 V, 20 mA) is connected to a 12 V supply. State why a series resistor is needed and calculate its value. (3)

[25]

QUESTION 5: LOGIC AND SWITCHING CIRCUITS

5.1 Use Boolean algebra to simplify $Y = A\cdot B + A\cdot\overline{B} + \overline{A}\cdot B$. Show every step. (5)

5.2 Draw the complete truth table for $Y = (A + B)\cdot\overline{C}$. (6)

5.3 Use a Karnaugh map to simplify $F(A, B, C) = \Sigma\, m(1, 3, 5, 6, 7)$, where A is the most significant bit. (6)

5.4 Explain briefly how a 555 timer in astable mode produces a square wave. Then calculate its frequency if $R_1 = 10\ k\Omega$, $R_2 = 47\ k\Omega$ and $C = 10\ \mu F$. Use $f = 1{,}44 \div ((R_1 + 2R_2) \times C)$. (5)

5.5 Distinguish between combinational and sequential logic circuits, with ONE example of a sequential circuit. (3)

[25]

TOTAL: 100

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END OF QUESTION PAPER